

Typing Rules and Pi Injectivity for Dependent Type Theory with $\mathbf{U} : \mathbf{U}$

1 Judgments

We work with three mutually defined judgment forms:

$\Gamma \text{ ctx}$	Γ is a well-formed context
$\Gamma \vdash M : A$	M has type A in context Γ
$\Gamma \vdash M = N : A$	M and N are definitionally equal at type A

Contexts are snoc-lists: $\Gamma ::= \cdot \mid \Gamma, x:A$.

2 Well-formed contexts

$$\frac{}{\cdot \text{ ctx}} \qquad \frac{\Gamma \vdash A : \mathbf{U}}{\Gamma, x:A \text{ ctx}}$$

3 Typing rules

$$\begin{array}{c} \text{VAR} \\ \frac{\Gamma \text{ ctx} \quad (x:A) \in \Gamma}{\Gamma \vdash x : A} \end{array} \qquad \begin{array}{c} \text{CONV} \\ \frac{\Gamma \vdash M : A \quad \Gamma \vdash A = B : \mathbf{U} \quad \Gamma \vdash B : \mathbf{U}}{\Gamma \vdash M : B} \end{array} \qquad \begin{array}{c} \mathbf{U} \\ \frac{\Gamma \text{ ctx}}{\Gamma \vdash \mathbf{U} : \mathbf{U}} \end{array}$$

$$\begin{array}{c} \text{PI} \\ \frac{\Gamma \vdash A : \mathbf{U} \quad \Gamma, x:A \vdash B : \mathbf{U}}{\Gamma \vdash \Pi(x:A)B : \mathbf{U}} \end{array} \qquad \begin{array}{c} \text{LAM} \\ \frac{\Gamma \vdash A : \mathbf{U} \quad \Gamma, x:A \vdash B : \mathbf{U} \quad \Gamma, x:A \vdash M : B}{\Gamma \vdash \lambda(x:A)M : \Pi(x:A)B} \end{array}$$

$$\begin{array}{c} \text{APP} \\ \frac{\Gamma \vdash A : \mathbf{U} \quad \Gamma, x:A \vdash B : \mathbf{U} \quad \Gamma \vdash f : \Pi(x:A)B \quad \Gamma \vdash a : A}{\Gamma \vdash f a : B[a]} \end{array}$$

4 Conversion rules

$$\begin{array}{c}
\text{REFL} \\
\frac{\Gamma \vdash M : A}{\Gamma \vdash M = M : A} \\
\\
\text{SYM} \\
\frac{\Gamma \vdash M = N : A}{\Gamma \vdash N = M : A} \\
\\
\text{TRANS} \\
\frac{\Gamma \vdash M = N : A \quad \Gamma \vdash N = P : A}{\Gamma \vdash M = P : A} \\
\\
\text{CONV-EQ} \\
\frac{\Gamma \vdash M = N : A \quad \Gamma \vdash A = B : \mathbb{U} \quad \Gamma \vdash B : \mathbb{U}}{\Gamma \vdash M = N : B} \\
\\
\text{BETA} \\
\frac{\Gamma \vdash A : \mathbb{U} \quad \Gamma, x:A \vdash B : \mathbb{U} \quad \Gamma, x:A \vdash M : B \quad \Gamma \vdash a : A}{\Gamma \vdash (\lambda(x:A)M) a = M[a] : B[a]} \\
\\
\text{PI-CONG} \\
\frac{\Gamma \vdash A = A' : \mathbb{U} \quad \Gamma, x:A \vdash B = B' : \mathbb{U}}{\Gamma \vdash \Pi(x:A)B = \Pi(x:A')B' : \mathbb{U}} \\
\\
\text{FUNEXT} \\
\frac{\Gamma \vdash A : \mathbb{U} \quad \Gamma, x:A \vdash f^+ x = g^+ x : B \quad \Gamma \vdash f : \Pi(x:A)B \quad \Gamma \vdash g : \Pi(x:A)B}{\Gamma \vdash f = g : \Pi(x:A)B} \\
\\
\text{APP-FUN} \\
\frac{\Gamma \vdash A : \mathbb{U} \quad \Gamma, x:A \vdash B : \mathbb{U} \quad \Gamma \vdash f = f' : \Pi(x:A)B \quad \Gamma \vdash a : A}{\Gamma \vdash f a = f' a : B[a]} \\
\\
\text{APP-ARG} \\
\frac{\Gamma \vdash A : \mathbb{U} \quad \Gamma, x:A \vdash B : \mathbb{U} \quad \Gamma \vdash f : \Pi(x:A)B \quad \Gamma \vdash a = a' : A}{\Gamma \vdash f a = f a' : B[a]}
\end{array}$$

Theorem 1 (Eta-conversion). *If $\Gamma \vdash A : \mathbb{U}$, $\Gamma, x:A \vdash B : \mathbb{U}$, and $\Gamma \vdash f : \Pi(x:A)B$, then*

$$\Gamma \vdash f = \lambda(x:A)f^+ x : \Pi(x:A)B.$$

Proof. By FUNEXT it suffices to show $\Gamma, x:A \vdash f^+ x = (\lambda(x:A)f^+ x)^+ x : B$. Since $(\lambda(x:A)M)^+ = \lambda(x:A^+)M^{+\ell}$ (where $+\ell$ denotes weakening under the binder), the right-hand side is a β -redex. Applying BETA and the substitution identity $M^{+\ell}[x_0] = M$ gives the result. Formalised in `EtaConversion.agda`. $\square$

5 Head reduction

We define head reduction $M \rightsquigarrow N$ as the compatible closure of β -reduction at the head:

$$\begin{array}{c}
\text{HR-BETA} \\
\frac{}{(\lambda(x:A)M) a \rightsquigarrow M[a]} \\
\\
\text{HR-APP} \\
\frac{M \rightsquigarrow M'}{M a \rightsquigarrow M' a}
\end{array}$$

6 Main result: Pi injectivity

Theorem 2 (Corollary 6; Coquand–Huber 2018, p.661). *If $\Gamma \vdash A_0 = \Pi(x:B_1)F_1 : \mathbb{U}$, then there exist B_0, F_0 such that*

1. $A_0 \rightsquigarrow^* \Pi(x:B_0)F_0$,
2. $\Gamma \vdash B_0 = B_1 : \mathbb{U}$,
3. $\Gamma, x:B_0 \vdash F_0 = F_1 : \mathbb{U}$.

Corollary 3 (Pi–Pi injectivity). *If $\Gamma \vdash \Pi(x:A_0)B_0 = \Pi(x:A_1)B_1 : \mathbb{U}$, then*

$$\Gamma \vdash A_0 = A_1 : \mathbb{U} \quad \text{and} \quad \Gamma, x:A_0 \vdash B_0 = B_1 : \mathbb{U}.$$

Proof outline (Theorem). The proof proceeds by domain-theoretic adequacy.

1. From $\Gamma \vdash A_0 = \Pi B_1 F_1 : \mathbb{U}$, extract $\Gamma \vdash A_0 : \mathbb{U}$ by presupposition (SUBSTITUTIONLEMMA).
2. Apply the soundness theorem (CONVSOUND') at the *bottom environment* $\rho_\perp$ (mapping every variable to $\perp$) to obtain a bidirectional evaluation transfer: for every finite approximation u ,

$$\text{EvalRel}(\Pi B_1 F_1, \rho_\perp, u) \implies \text{EvalRel}(A_0, \rho_\perp, u).$$

3. Instantiate with $u = \text{PiCode}(\perp, \text{nil})$, which is coherent because $\text{CoherentFunTail}(\text{nil}) = \top$. The evaluation $\text{EvalRel}(\Pi B_1 F_1, \rho_\perp, \text{PiCode}(\perp, \text{nil}))$ is constructible trivially (domain $\perp$, empty function graph, vacuous selection body). Transfer gives $\text{EvalRel}(A_0, \rho_\perp, \text{PiCode}(\perp, \text{nil}))$.
4. Apply the bundled equality adequacy theorem (ADEQUACYEQSUB2) with the identity substitution at $\rho_\perp$ to obtain

$$\text{EqVal2 } \Gamma \ A_0 \ (\Pi B_1 F_1) \ \mathbb{U} \ (\text{PiCode}(\perp, \text{nil})) \ \text{UCode}.$$

5. This unfolds to EqValTyPi2 , a Σ -type containing:

- $\text{Red } \Gamma \ A_0 \ (\Pi B_0 F_0) \ \mathbb{U}$ (giving part 1),
- $\Gamma \vdash B_0 = B'_1 : \mathbb{U}$ and $\Gamma, x:B_0 \vdash F_0 = F'_1 : \mathbb{U}$,
- $\text{Red } \Gamma \ (\Pi B_1 F_1) \ (\Pi B'_1 F'_1) \ \mathbb{U}$.

Since Π is a head normal form, Red-unique-Pi forces $B'_1 = B_1$ and $F'_1 = F_1$, giving parts 2 and 3. □

Proof of Corollary. Apply the theorem to $\Gamma \vdash \Pi A_0 B_0 = \Pi A_1 B_1 : \mathbb{U}$. This yields B'_0, F'_0 with $\Pi A_0 B_0 \rightsquigarrow^* \Pi B'_0 F'_0$. Since Π is a head normal form, Red-unique-Pi gives $B'_0 = A_0$ and $F'_0 = B_0$, so the domain and codomain conversions become $\Gamma \vdash A_0 = A_1 : \mathbb{U}$ and $\Gamma, x:A_0 \vdash B_0 = B_1 : \mathbb{U}$. □

The entire proof is formalised in Agda with `--without-K` `--exact-split`, no `--type-in-type`, and no postulates (in the files on which `PiInjectivity.agda` depends).

References

- [1] T. Coquand and S. Huber, “An adequacy theorem for dependent type theory,” *Archive for Mathematical Logic*, vol. 57, pp. 649–676, 2018.